1d





AT $x=0$ $\quad J(x=0) = J_0$

$c(x=0) = c_0$ BC 1

i AT STEADY STATE, WHAT IS $c(x)$?

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

D.E.

$$\frac{\partial c}{\partial t} = 0 \qquad D \frac{\partial^2 c}{\partial x^2} = 0$$

$$J = -D \frac{\partial c}{\partial x}$$

AT $x=0$, $\quad J_0 = -D \frac{\partial c}{\partial x}$

$$\Rightarrow \frac{\partial c}{\partial x} = -\frac{J_0}{D}$$ BC 2

ii ASSUME CONSTANT CONSUMPTION

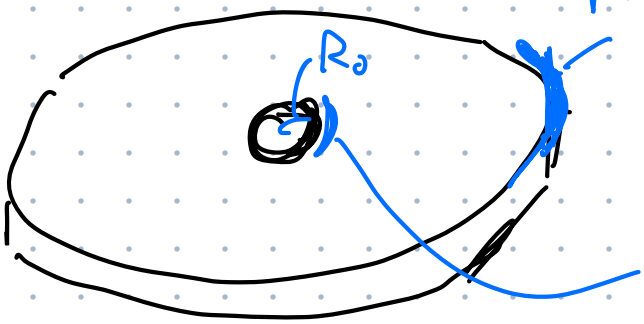
$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} - E$$

AT STEADY STATE /

$$D\frac{\partial^2 c}{\partial x^2} - E = 0$$

D.E.

PS 3



$J = J_0$ OUTWARD

$c = c_0$

i) $J(R_{MAX}) =$

$c(r) =$

$$c(r) = c_1 \ln r + c_2$$

$$J(R_0) = -D\frac{c_1}{R_0} = J_0$$

$$c(r) = \frac{R_0 J_0}{D} \ln\left(\frac{R_0}{r}\right) + c_0 \quad \checkmark$$

CHECK $r = R_0$,

$c = c_0$ ✓

ii) $c(r) \stackrel{?}{=} \frac{1}{D}\left(\frac{r^2 \chi}{4} - R_0 J_0 \ln\left(\frac{r}{R_0}\right)\right)$

$+ c_0$